Day 8 Problem Set

Geometry Summer Credit | Similarity and Proportions

Name:	Date:	Period:

Goal

Practice scale factor, dilations, AA similarity, and proportions from similar figures.

Support Practice

1.	Solve the proportion: 2/3 = x/18.

2.	Solve the proportion: 5/x = 10/24.

3.	Dilate A(3, -4) by scale factor 2 about the origin.

4.	A side of length 11 is scaled by factor 3. Find the new length.

5.	A side of length 30 is the image after scale factor 5. Find the original length.

Core Practice: Scale Factor and Dilations

6.	A triangle has side lengths 4, 6, 9. A similar triangle has side lengths 12, 18, 27. Find the scale factor from the first triangle to the second.

7.	A rectangle has side lengths 5 and 8. A similar rectangle has side lengths 20 and 32. Find the scale factor.

8.	A dilation maps (x, y) to (3x, 3y). Find the image of P(-2, 7).

9.	A dilation maps R(12, -8) to R'(3, -2). What is the scale factor?

10.	A map uses 1 cm = 8 km. Two towns are 6 cm apart on the map. How far apart are they in real life?

11.	Error analysis: A student says a dilation by scale factor 2 creates a congruent figure. Correct the statement.

Day 8 Problem Set

Continued

Core Practice: AA Similarity and Missing Sides

12.	Triangle ABC has angles 35 and 80 degrees. Triangle DEF has angles 80 and 35 degrees. Are the triangles similar? Explain.

13.	Triangle ABC is similar to triangle DEF. AB = 7, DE = 21, BC = 10. Find EF.

14.	Triangle RST is similar to triangle XYZ. RS = 12, XY = 30, ST = 16. Find YZ.

15.	Triangle JKL is similar to triangle MNP. JK = 9, MN = 15, NP = 20, KL = x. Find x.

16.	Triangle ABC is similar to triangle XYZ. AB = 8, XY = 20, AC = 12, XZ = y. Find y.

17.	Solve: 6/15 = 10/x.

18.	Solve: x/18 = 14/21.

19.	A 5-foot student casts a 7-foot shadow. A flagpole casts a 28-foot shadow. Use similarity to find the flagpole height.

20.	Spiral: A rectangle has diagonals 2x + 3 and 25. Find x.

21.	Challenge: Scale factor 3. What happens to perimeter? What happens to area?